

Machine-Failure Model Comparison

Table 1 – 5-fold CV (80% train)

Model	Best Params	CV MCC (5-fold)
MLP	{'clfllearning_rate': 'constant', 'clfhidden_layer_sizes': (100, 50), 'clfalphabet': 0.001, 'clfactivation': 'relu'}	0.8232
SVM	{'clfkernell': 'rbf', 'clfgammat': 'auto', 'clf__C': 100}	0.8196
KNN	{'clfp': 2, 'clfn_neighboars': 7, 'clf__algorithm': 'ball_tree'}	0.7539
DecisionTree	{'clfccp_alphabet': 0.0, 'clfcriterion': 'entropy', 'clf__max_depth': 5}	0.8052
LogisticRegression	{'clfC': 10, 'clfpentalty': 'l2'}	0.632

Table 2 – Hold-out Test (20%)

Model	Best Params	Test MCC (20%)
MLP	{'clfllearning_rate': 'constant', 'clfhidden_layer_sizes': (100, 50), 'clfalphabet': 0.001, 'clfactivation': 'relu'}	0.8239
SVM	{'clfkernell': 'rbf', 'clfgammat': 'auto', 'clf__C': 100}	0.7945
KNN	{'clfp': 2, 'clfn_neighboars': 7, 'clf__algorithm': 'ball_tree'}	0.7945
DecisionTree	{'clfccp_alphabet': 0.0, 'clfcriterion': 'entropy', 'clf__max_depth': 5}	0.747
LogisticRegression	{'clfC': 10, 'clfpentalty': 'l2'}	0.6062

Conclusion

The **MLP** achieved the highest MCC on both CV and the test set, so it is selected as the production model.